



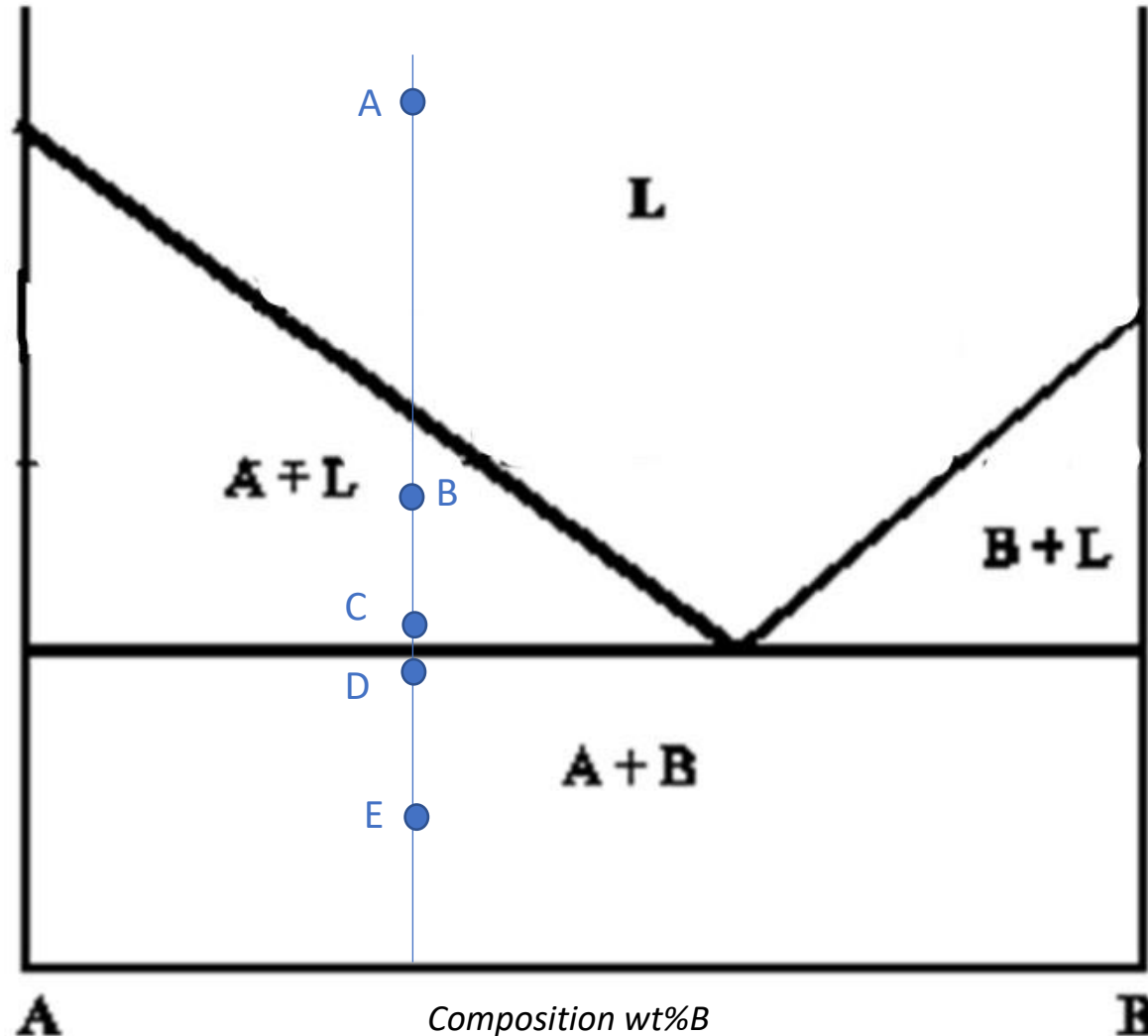
STM Exercise 6

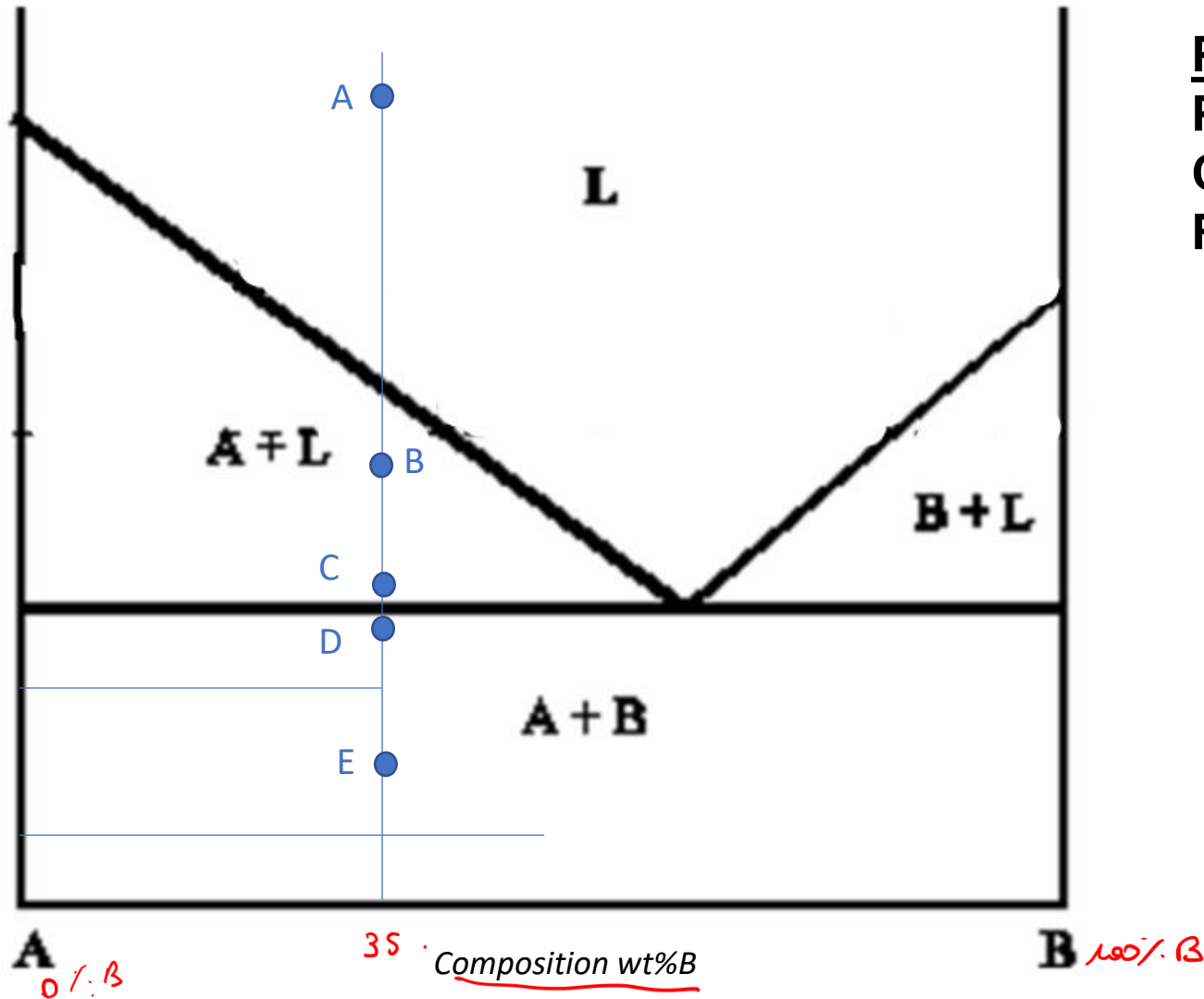
Phase Diagrams

(no solubility in the solid state)

(partial solubility in the solid state)

Exercise 1: determine the phase present, their composition and relative amounts in the point **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.





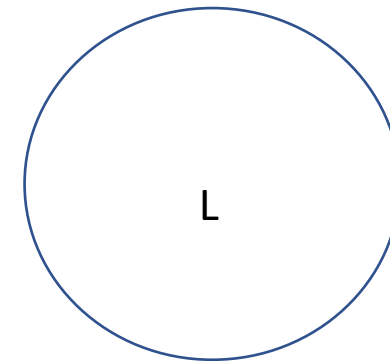
Point A:

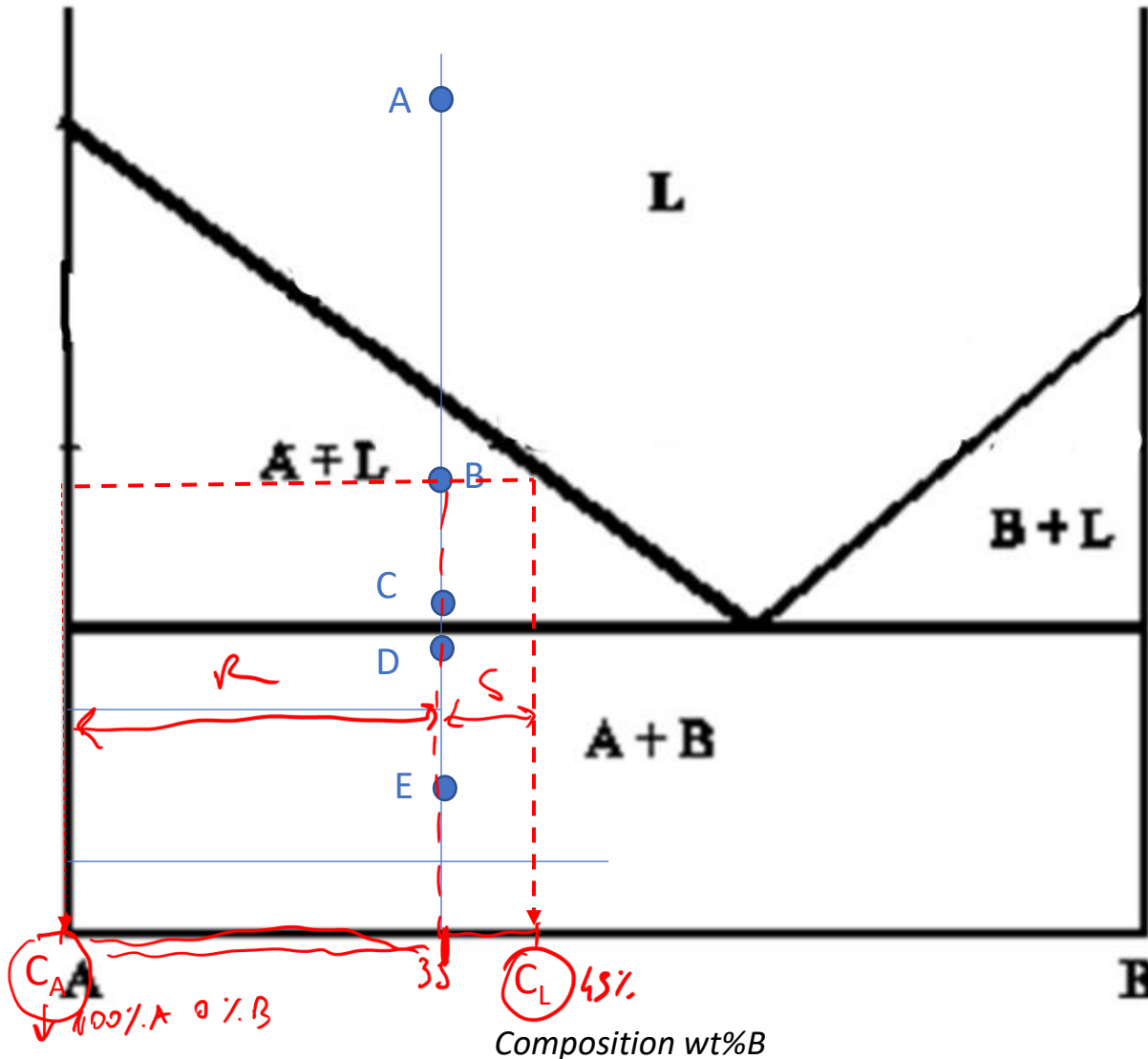
Phases: 1 phase liquid

Composition: 35%B, 65%A

$$100 - 35 = 65\% \text{ A}$$

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (A) + liquid (L)

Composition: 2 phases region → horizontal rule

Composition of liquid (L): $C_L = 45\% \text{ B}$, $55\% \text{ A}$ $(100 - 45) = 55\% \text{ A}$

Composition of solid (A): $C_A = 100\% \text{ A}$

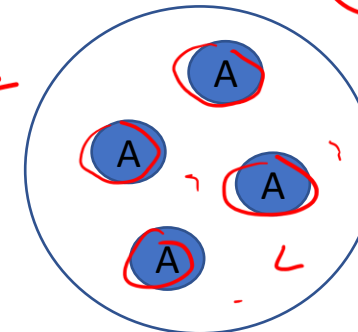
Relative amount: 2 phases region → lever rule

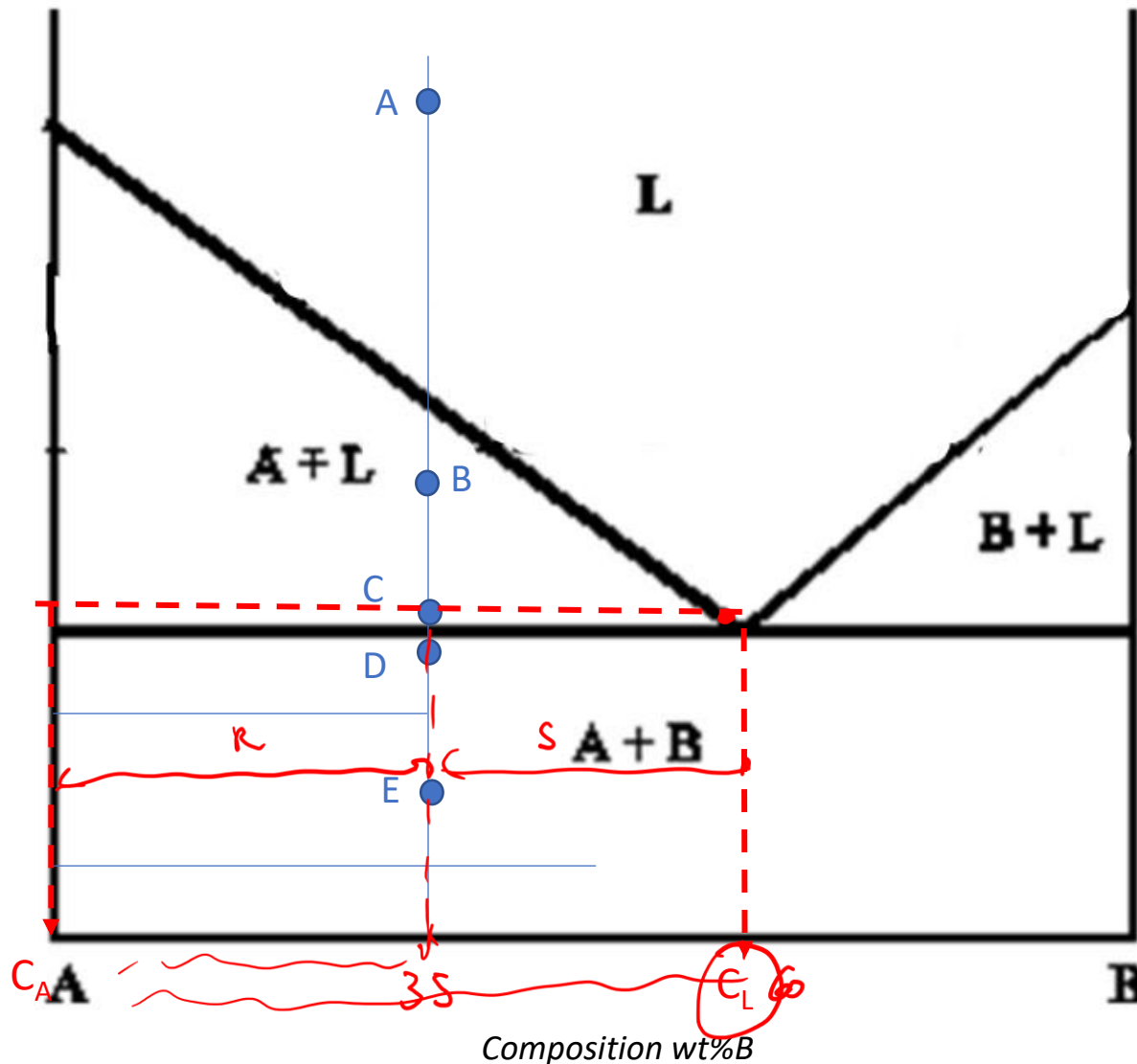
$$\% \text{ A} = (45 - 35) / (45 - 0) * 100 = 22\%$$

$$\% \text{ L} = (35 - 0) / (45 - 0) * 100 = 78\%$$

$$\% \text{ A} = \frac{S}{R+S} = \frac{45-35}{45}$$

$$\% \text{ L} = \frac{R}{R+S} = \frac{35}{45}$$





Point C:

Phases: 2 phases solid (A) + liquid (L)

Composition: 2 phases region → horizontal rule

Composition of liquid (L): $C_L = 60\% \text{ B}$, $40\% \text{ A}$

Composition of solid (A): $C_A = 100\% \text{ A}$

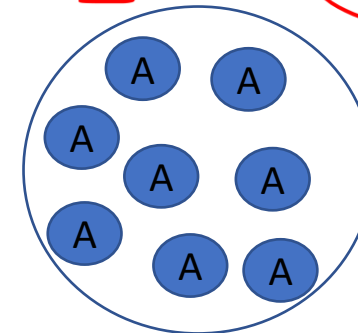
Relative amount: 2 phases region → lever rule

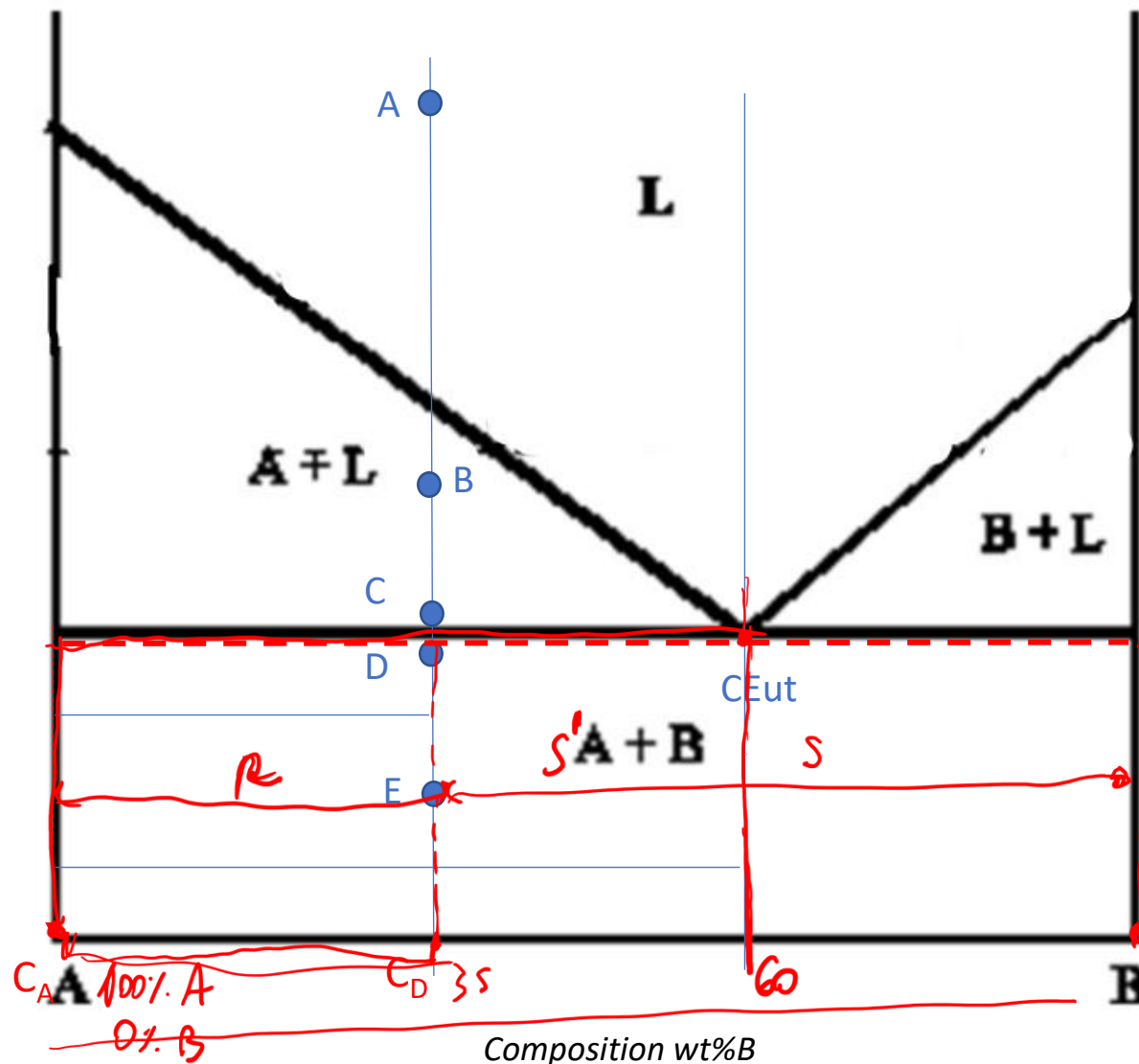
$$\% \text{ A} = \frac{(60-35)}{(60-0)} \times 100 = 42\%$$

$$\% \text{ L} = \frac{(35-0)}{(60-0)} \times 100 = 58\%$$

$$\% \text{ A} = \frac{S}{R+S}$$

$$\% \text{ L} = \frac{R}{R+S}$$





Point D:

Phases: 2 phases solid (A) + solid (B)

Composition: 2 phases region → horizontal rule

Composition of Solid (A): $C_A = 100\%A$

Composition of solid (B): $C_B = 100\%B$

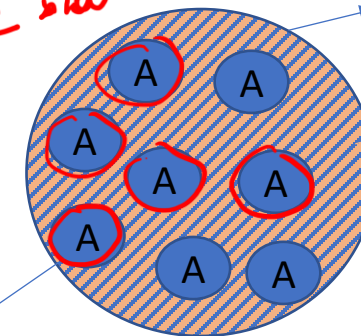
Relative amount: 2 phases region → lever rule

→ $\% A_{TOT} = (100-35)/(100-0) \times 100 = 65\%$

$\% B = (35-0)/(100-0) \times 100 = 35\%$

$\% A = \frac{S}{R+S} = \frac{100-35}{100} \times 100$

$\% B = \frac{R}{R+S} = \frac{35}{100}$



Eutectic A

Eutectic B

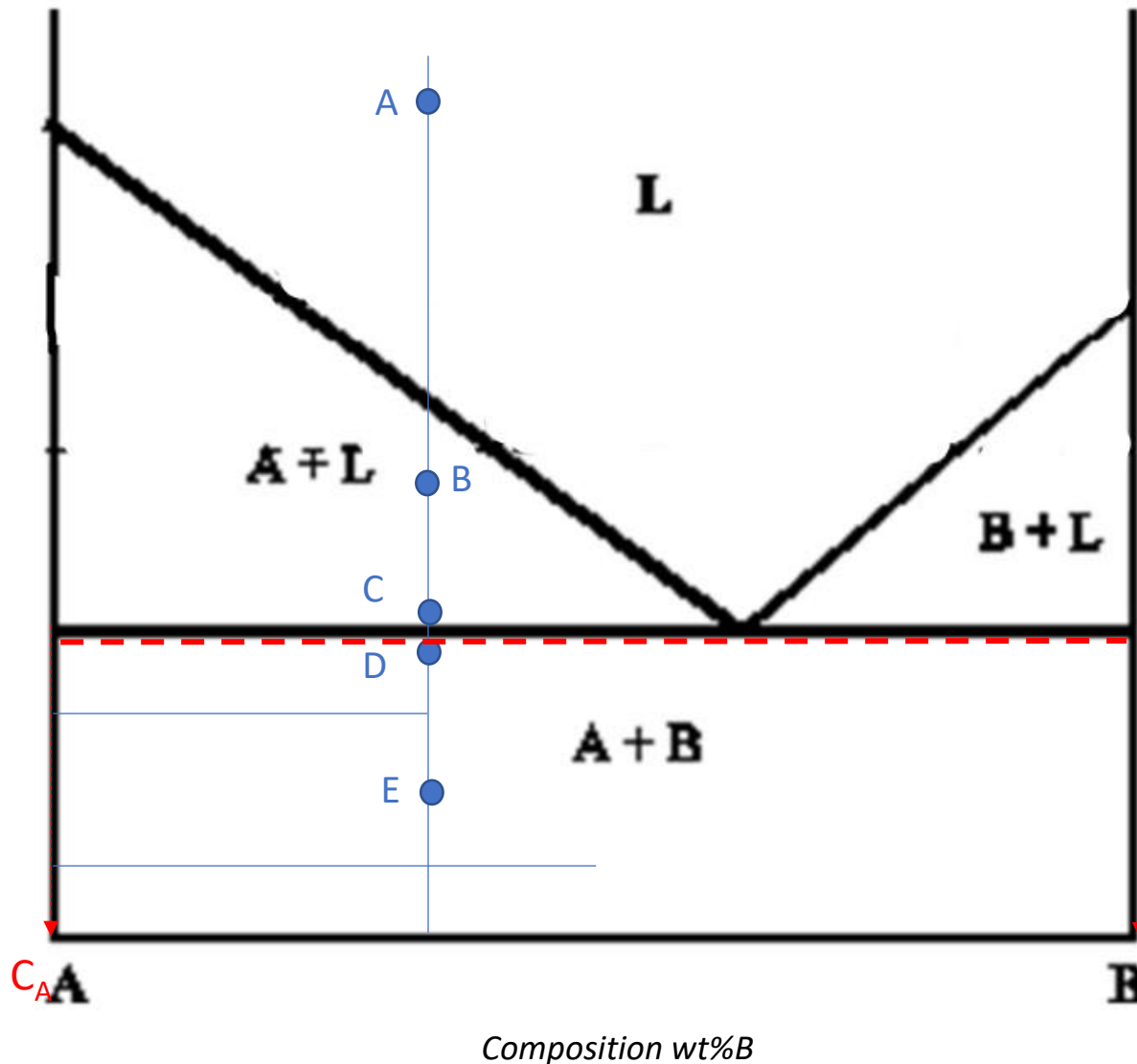
$\% A_{pr} = \frac{S'}{R+S'} = \frac{60-35}{60} \times 100 = 42\%$

$\% A_{EUT} = 65\% - 42\% = 23\%$

$\% \alpha_{primary} = (CEut - CD) / (CEut - CA) \times 100$ lever rule → CEut composition at eutectic reaction the last point where the primary solidification is finished. Using this value into the lever rule to find the primary A solid phase:

Primary/pro-eutectic A (As in point C but no formed anymore after eutectic): $\% A_{primary} = (60-35)/(60-0) \times 100 = 42\%$

$\% A_{EUT} = 65 - 42 = 23\%$ in (a+b)



Point E:

Phases: 2 phases solid (A) + solid (B) ✓

Composition: 2 phases region →
horizontal rule

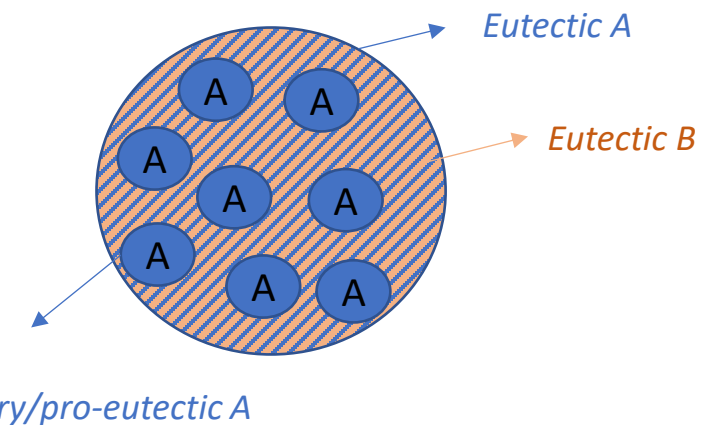
Composition of Solid (A): $C_A = 100\%A$ ✓

Composition of solid (B): $C_B = 100\%B$ ✓

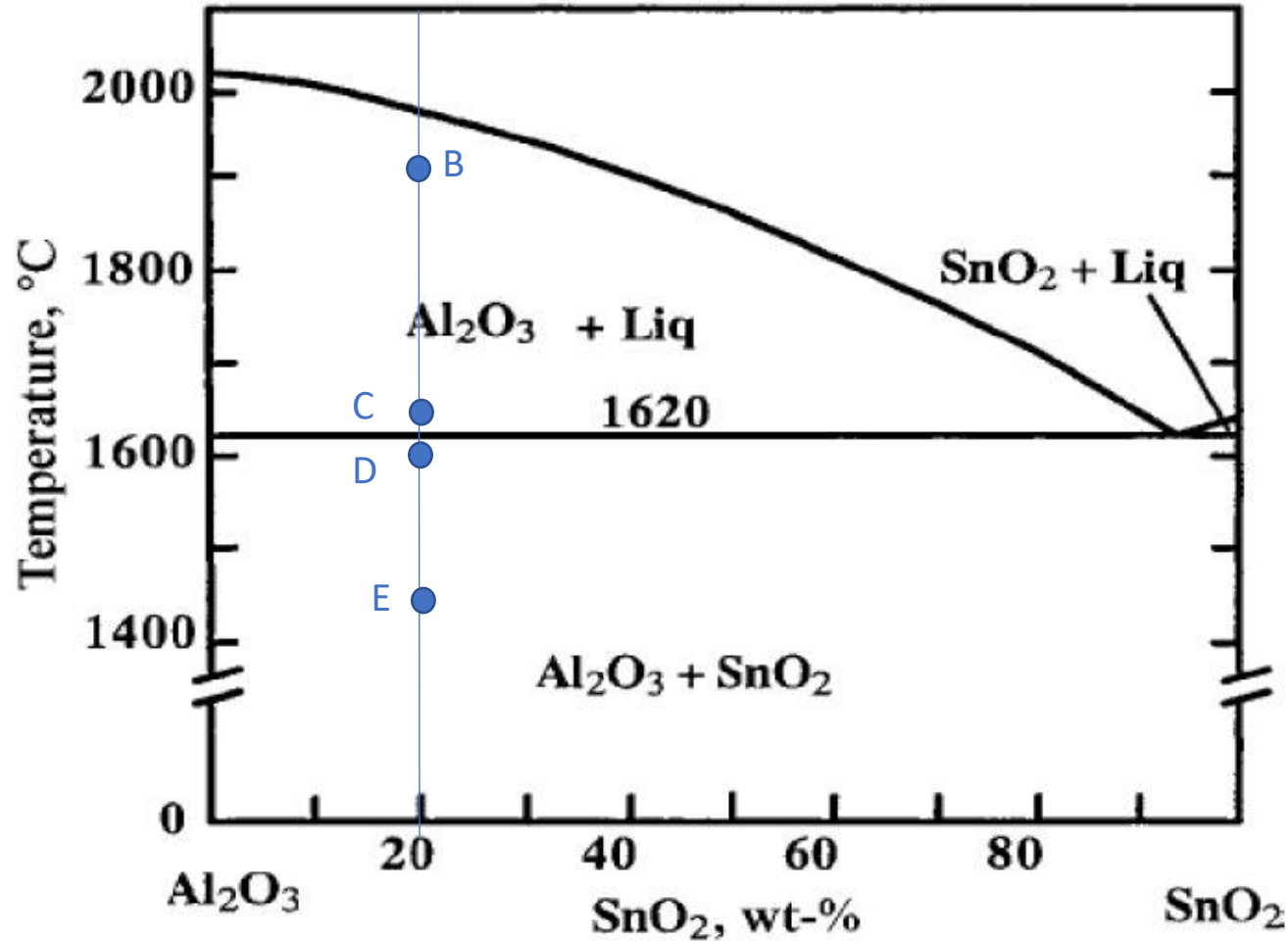
Relative amount: 2 phases region →
lever rule

$\%A = (100-35)/(100-0)*100=65\%$ ✓

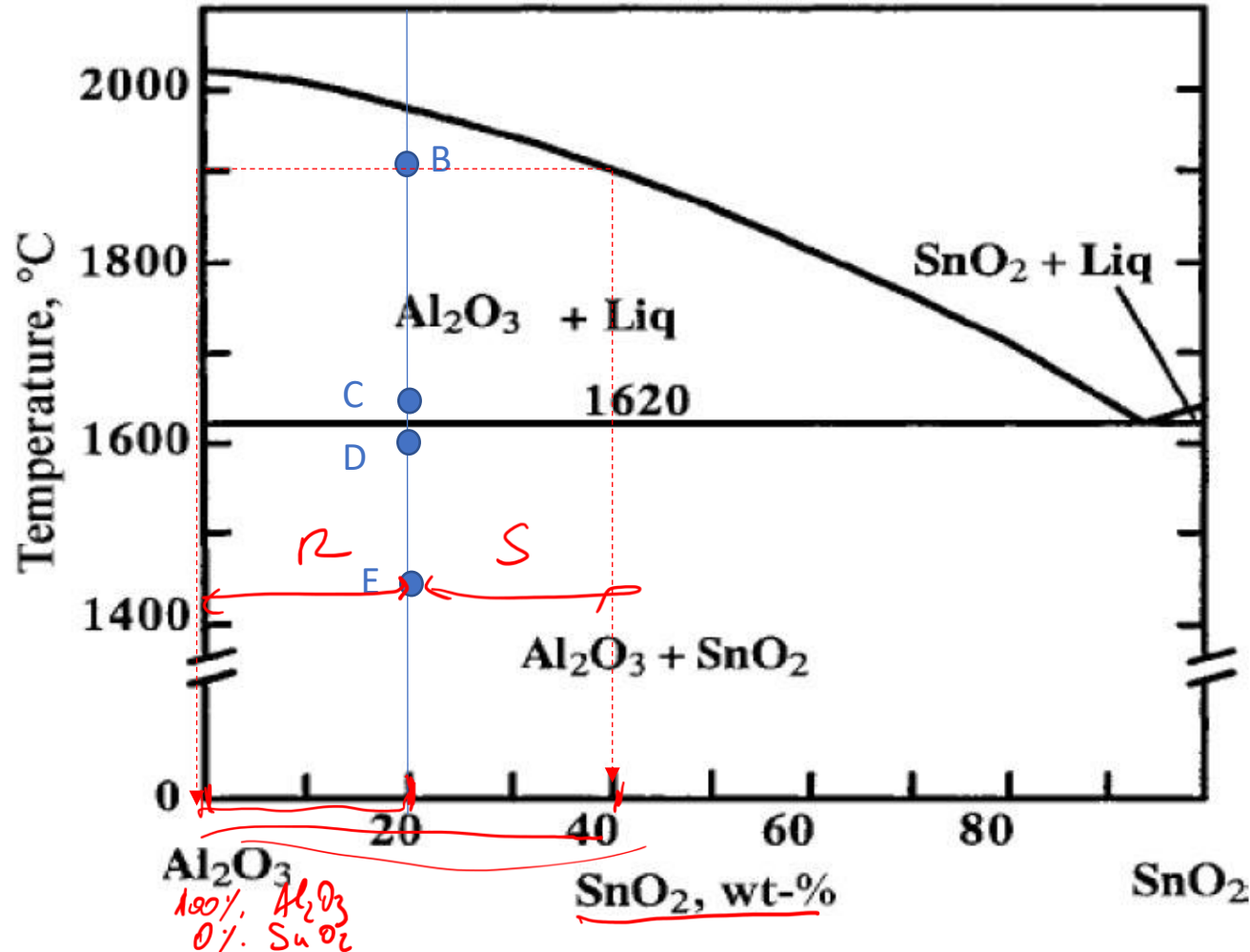
$\%B = (35-0)/(100-0)*100=35\%$ ✓



Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Point B:

Phases: 2 phases solid (Al_2O_3) + liquid (L)

Composition: 2 phases region → horizontal rule

Composition of liquid (L): = 40% SnO_2 , 60% Al_2O_3 *100 - 40 = 60%*

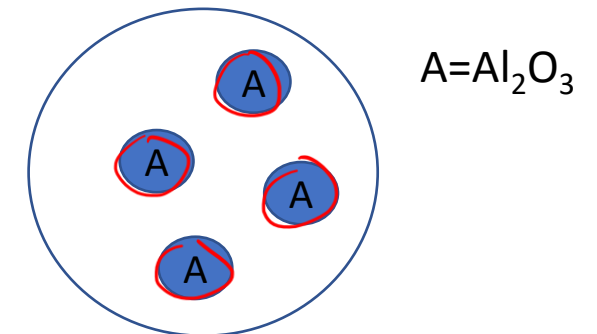
Composition of solid: = 100% Al_2O_3

Relative amount: 2 phases region → lever rule *S / R+S*

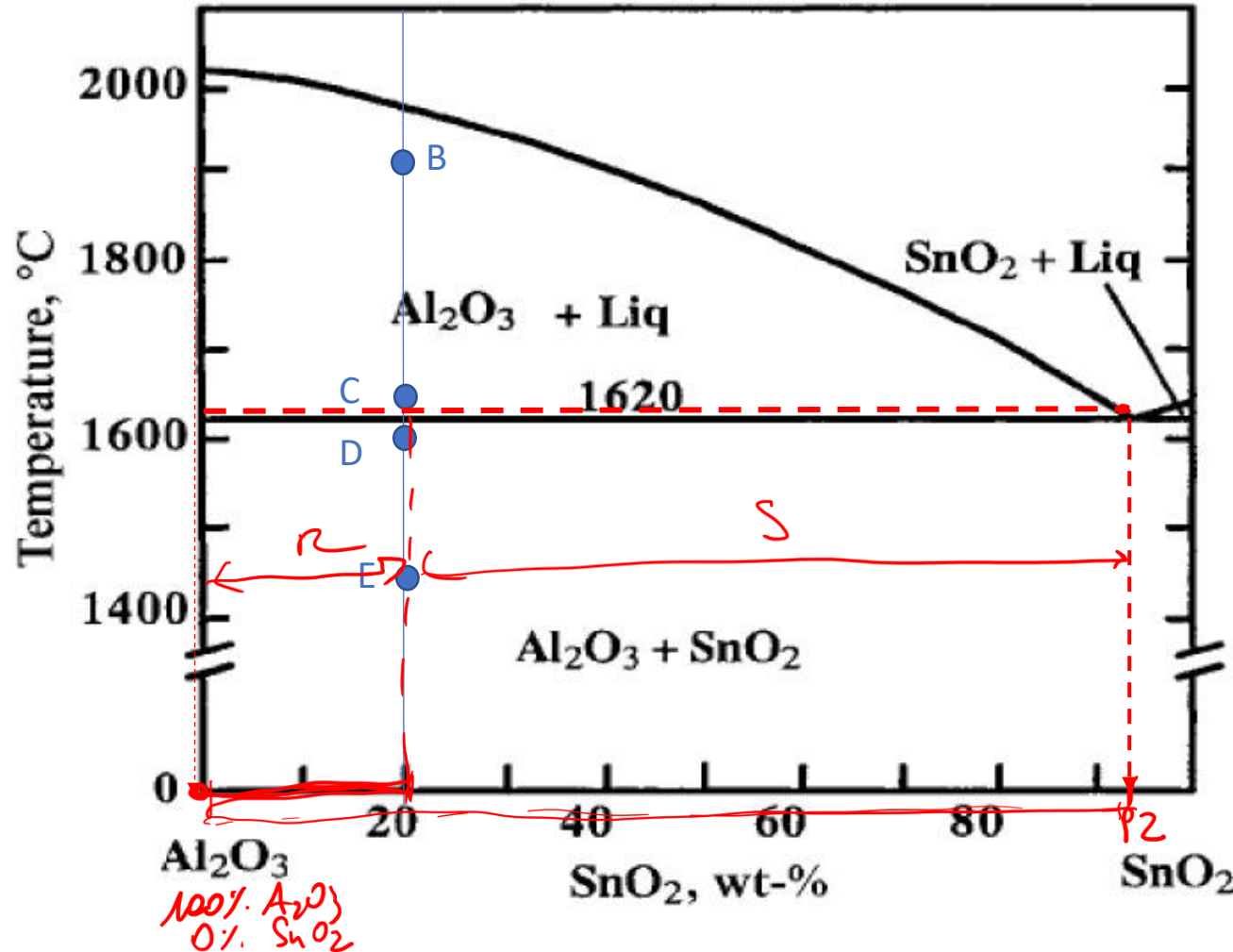
$$\% \text{Al}_2\text{O}_3 = \frac{(40-20)}{(40-0)} \times 100 = 50\%$$

$$\% \text{L} = \frac{(20-0)}{(40-0)} \times 100 = 50\%$$

R / R+S



Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Point C:

Phases: 2 phases solid (Al₂O₃) + liquid (L)

Composition: 2 phases region → horizontal rule

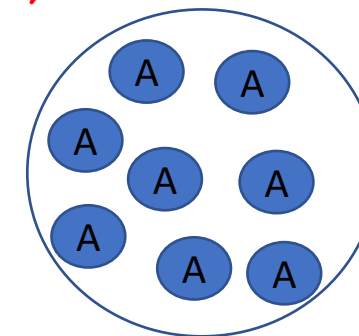
Composition of liquid (L): = 92% SnO₂, 8% Al₂O₃ (~~100~~ - 92) = 8

Composition of solid: = 100% Al₂O₃

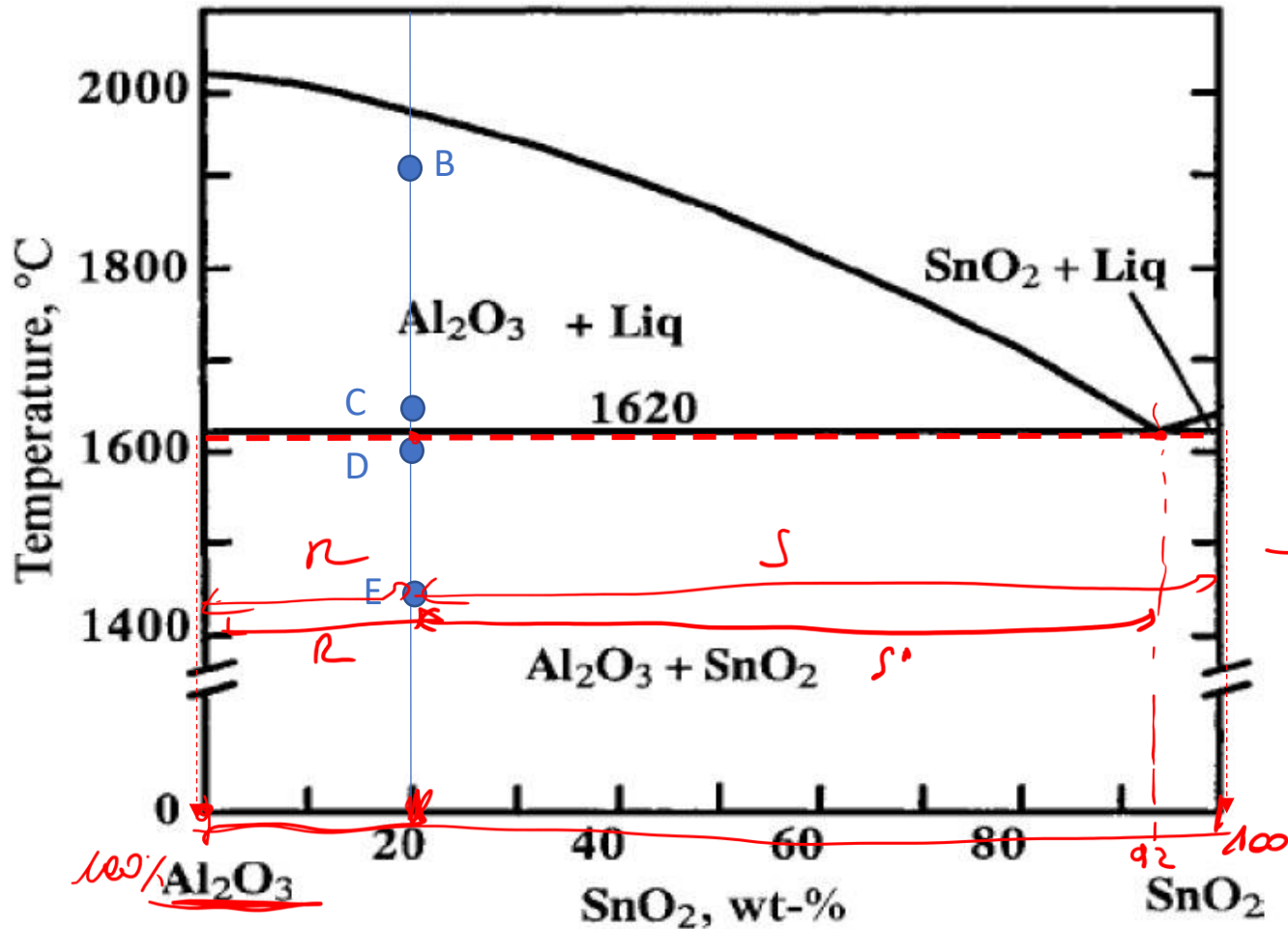
Relative amount: 2 phases region → lever rule

$$\% \text{Al}_2\text{O}_3 = \frac{(92-20)}{(92-0)} \times 100 = 78\%$$

$$\% \text{L} = \frac{(20-0)}{(92-0)} \times 100 = 22\%$$



A = Al₂O₃



Point D:

Phases: 2 phases solid (Al_2O_3) + solid (SnO_2)

Composition: 2 phases region $\rightarrow$ horizontal rule

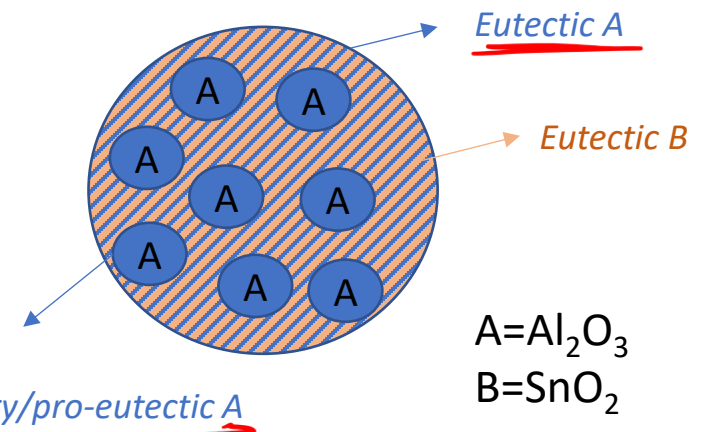
Composition of Solid A = $100\% \text{Al}_2\text{O}_3$

Composition of solid (B): $C_B = 100\% \text{SnO}_2$

Relative amount: 2 phases region $\rightarrow$ lever rule

$$\% \text{Al}_2\text{O}_3_{\text{TOT}} = \frac{(100-20)}{(100-0)} \times 100 = 80\%$$

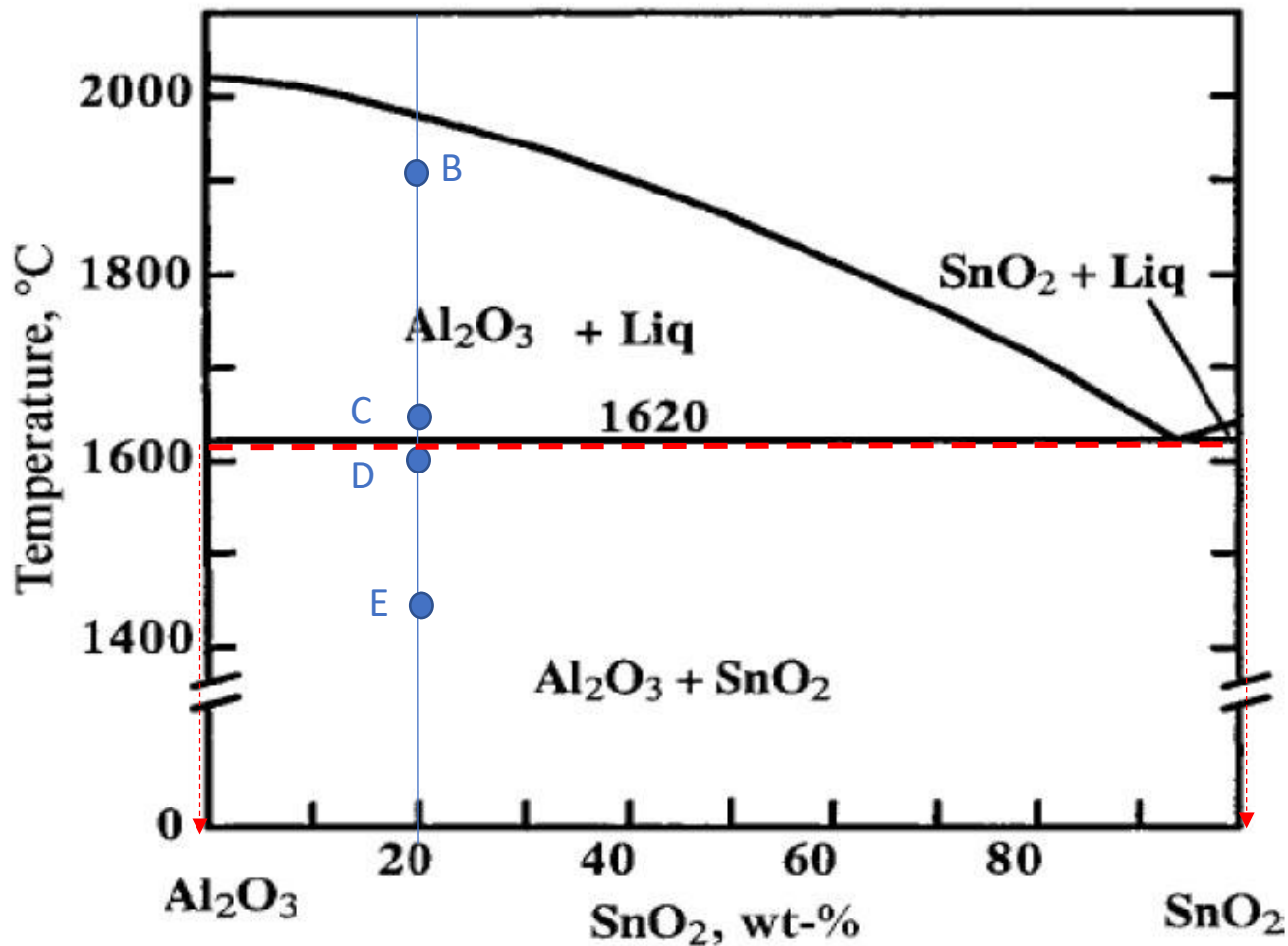
$$\% \text{SnO}_2 = \frac{(20-0)}{(100-0)} \times 100 = 20\%$$



Primary/pro-eutectic Al_2O_3 (As calculated in C, no more formed later): %

$$\text{Al}_2\text{O}_3_{\text{PRIM}} = \frac{(92-20)}{(92-0)} \times 100 = 78\%$$

$$\%A_{\text{EUT}} = 80 - 78 = 2\%$$



Point E:

Phases: 2 phases solid (Al_2O_3) + solid (SnO_2)

Composition: 2 phases region → horizontal rule

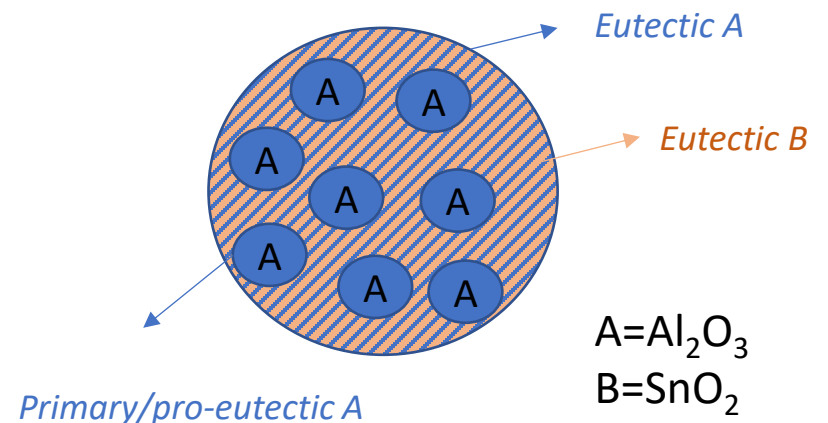
Composition of Solid A = 100% Al_2O_3

Composition of solid (B): $C_B = 100\%\text{SnO}_2$

Relative amount: 2 phases region → lever rule

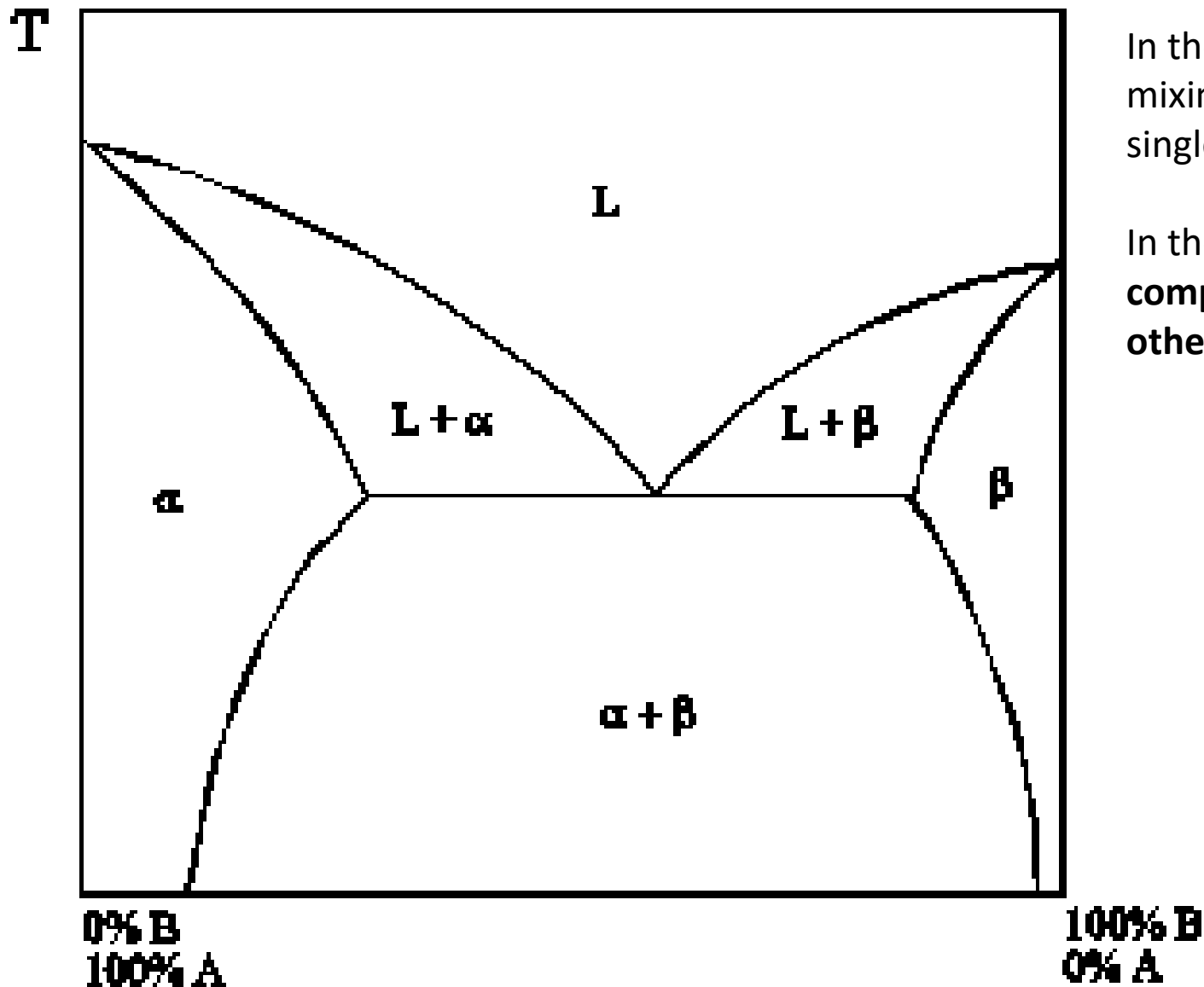
$$\% \text{Al}_2\text{O}_3_{\text{TOT}} = (100-20)/(100-0) \times 100 = 80\%$$

$$\% \text{SnO}_2 = (20-0)/(100-0) \times 100 = 20\%$$



Exercise 3: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **partial solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Complete solubility in the liquid state, partial solubility in the solid state



In the liquid state, A and B mixing $\rightarrow$ the system always forms single liquid phase (L).

In the **solid state**, A and B **do not completely dissolve** into each other

Complete solubility in the liquid state, partial solubility in the solid state

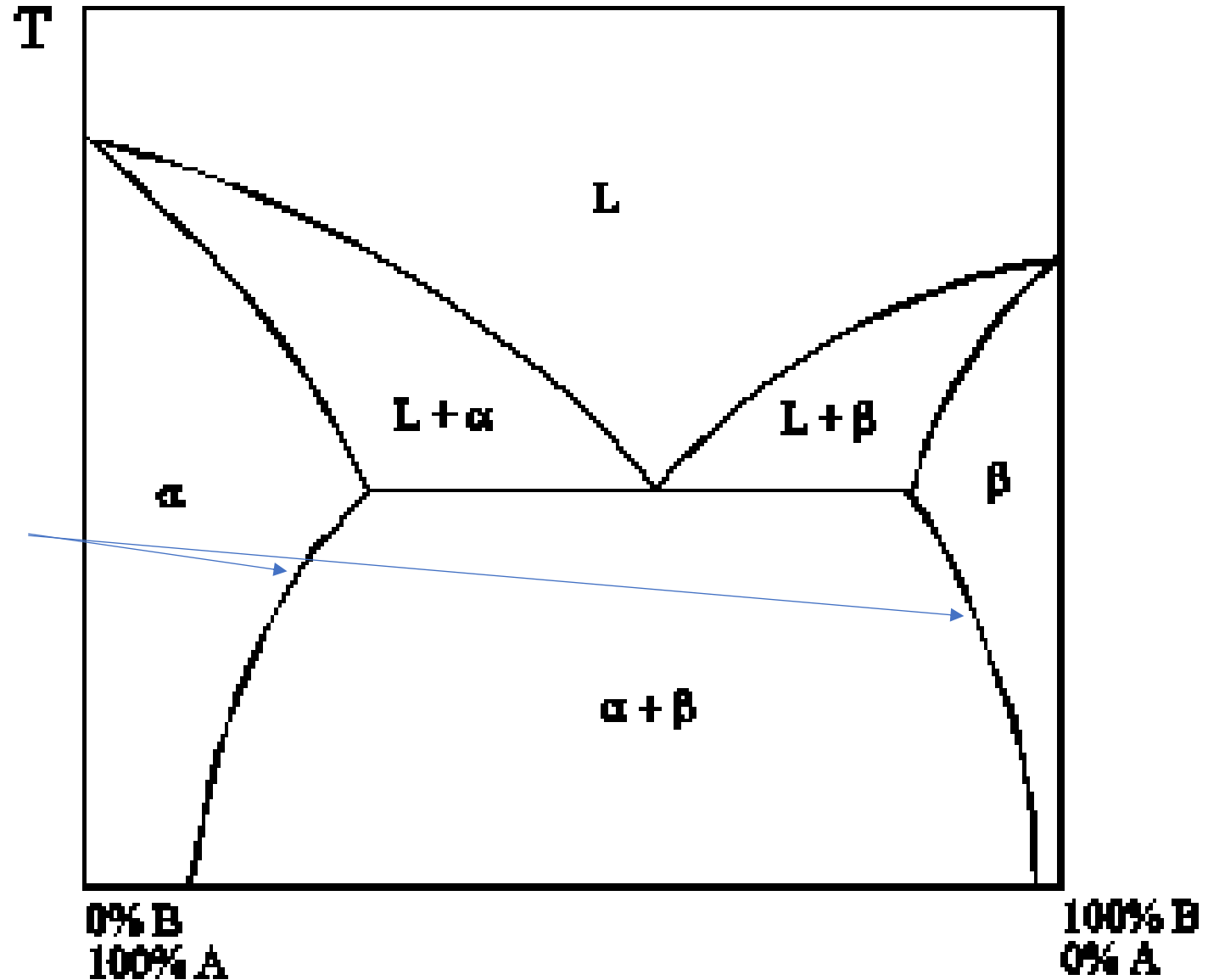
α Phase on the left

It is a solid solution rich in A

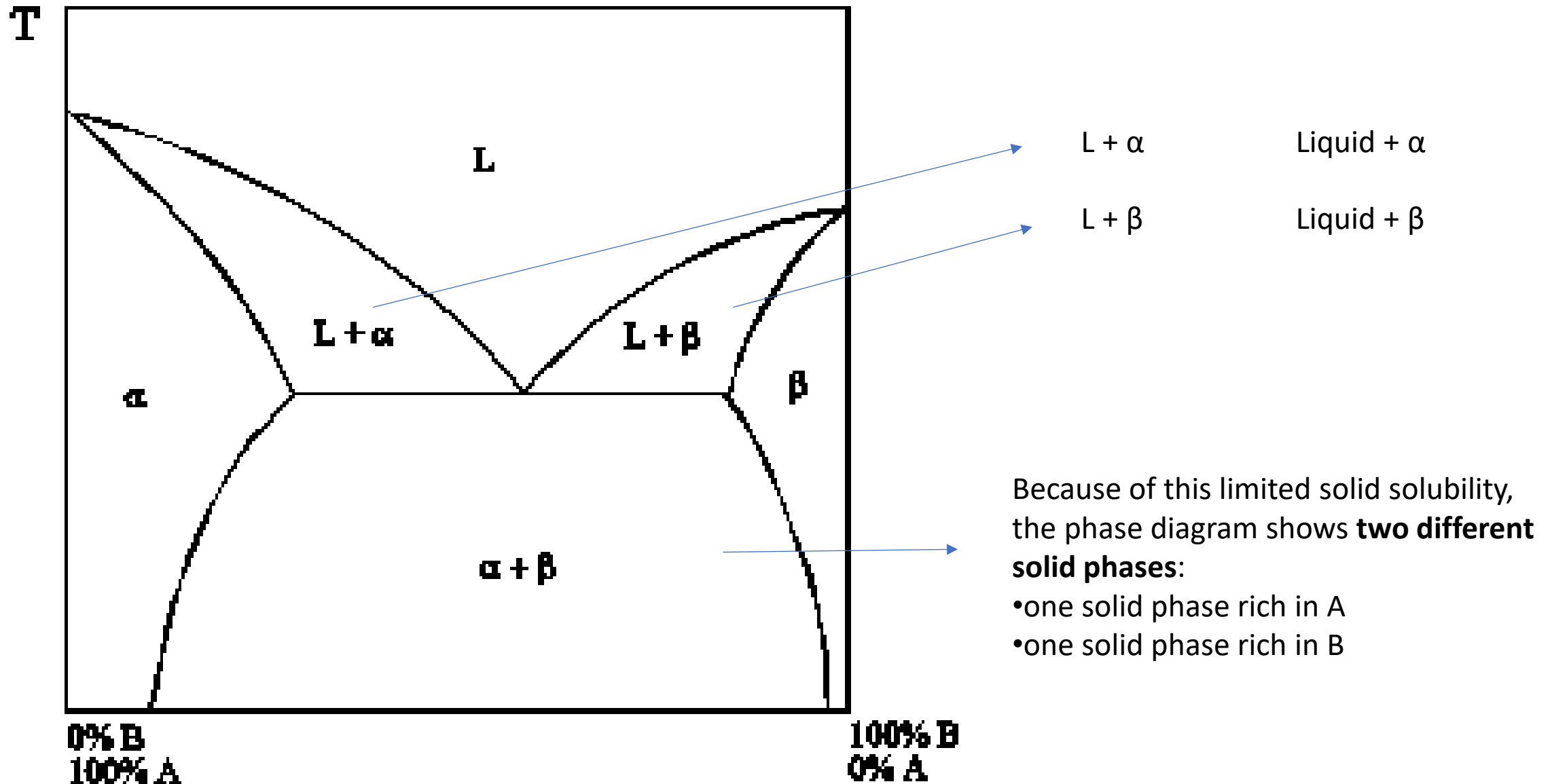
β PHASE :

- It is a solid solution rich in B

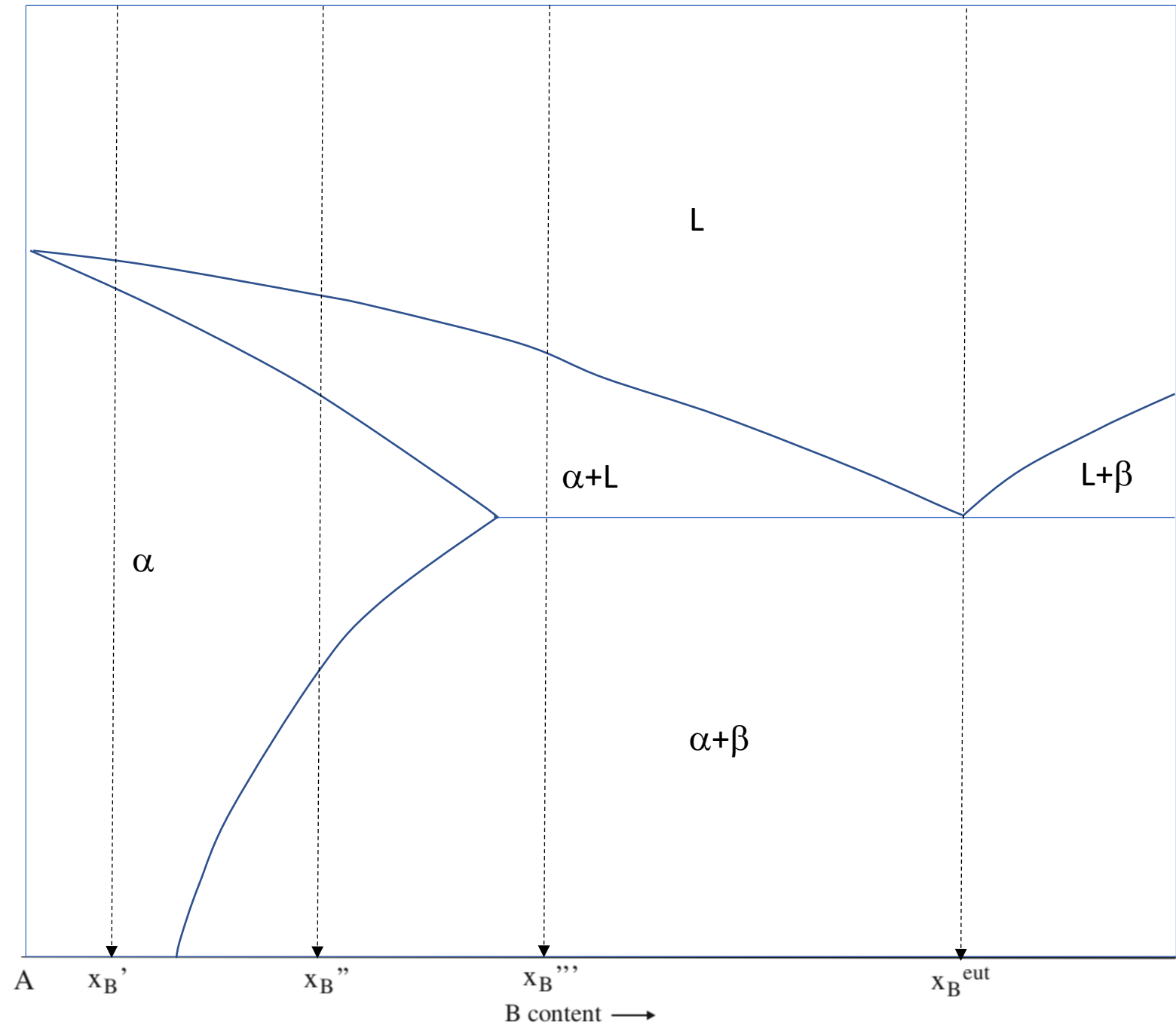
The solvus curve separates the region where a single solid solution exists

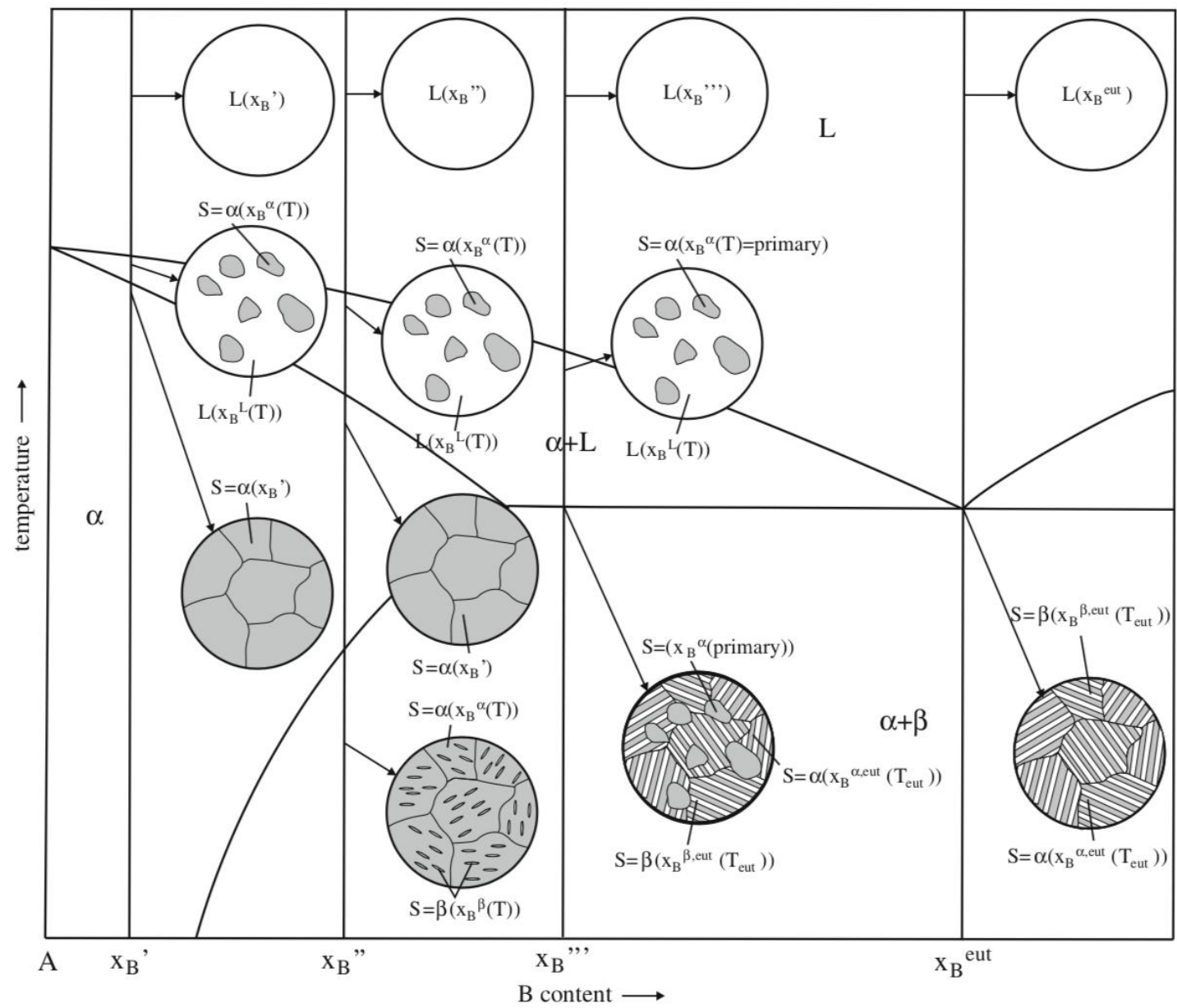


Complete solubility in the liquid state,
partial solubility in the solid state

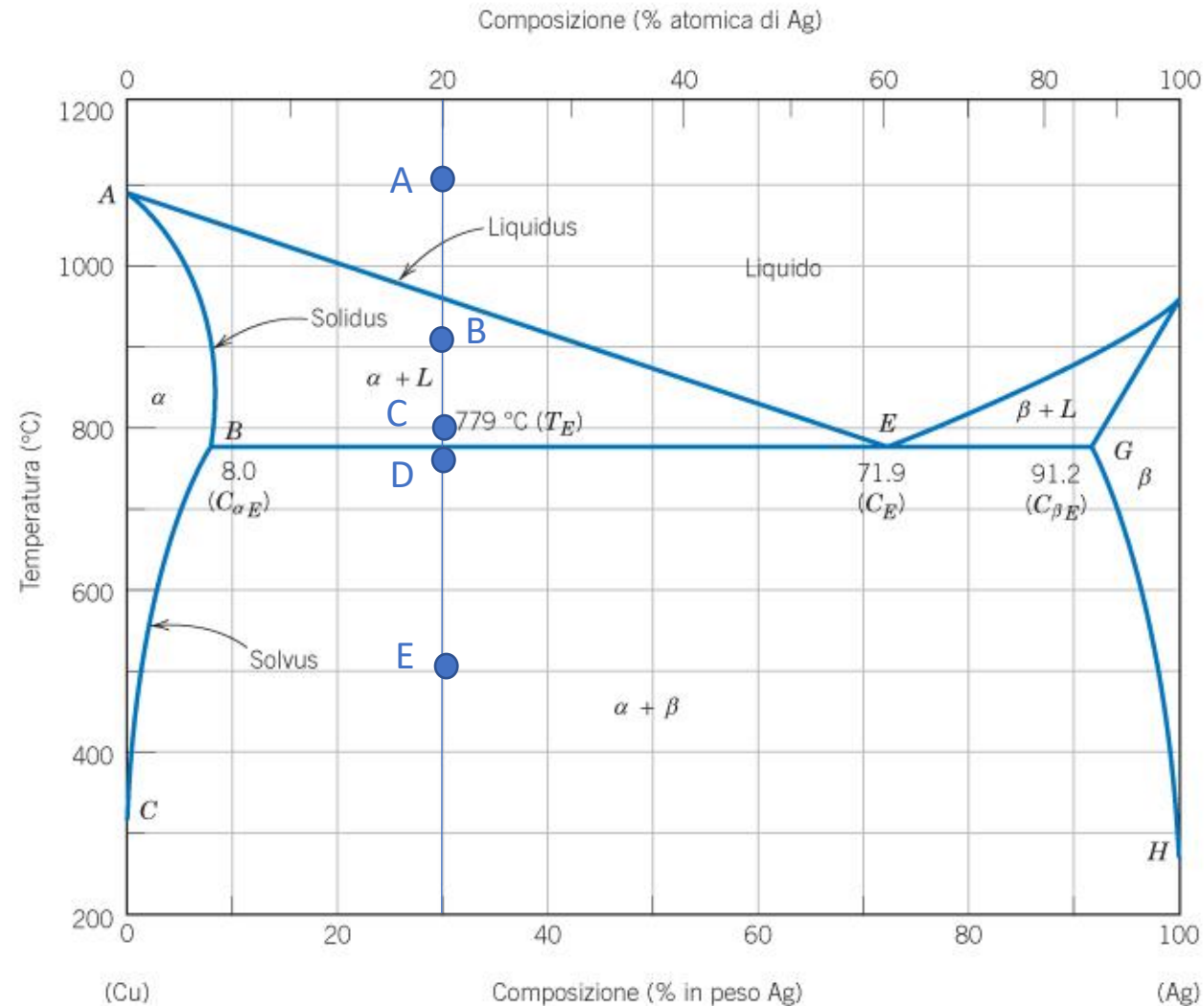


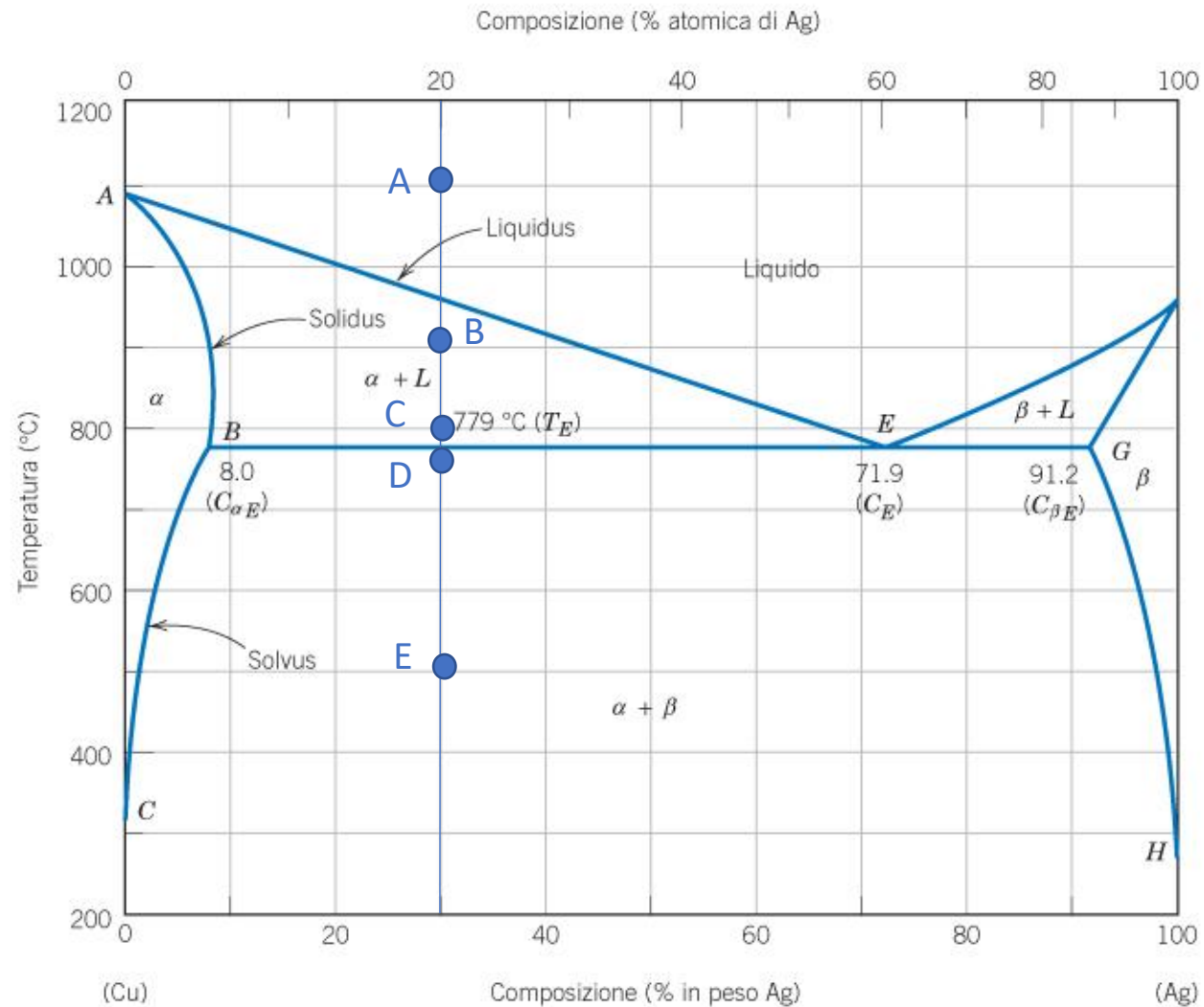
Exercise 4: describe the solidification of alloys with compositions x_B' , x_B'' , x_B''' and x_B^{eut} by drawing schematic microstructures in significant points.





Exercise 5: determine the phase present, their composition and relative amounts in the points **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



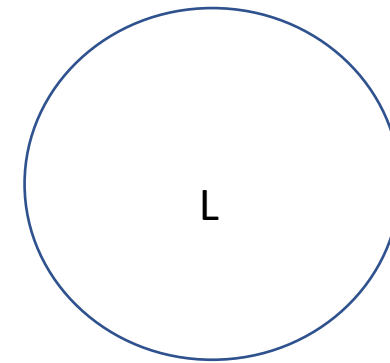


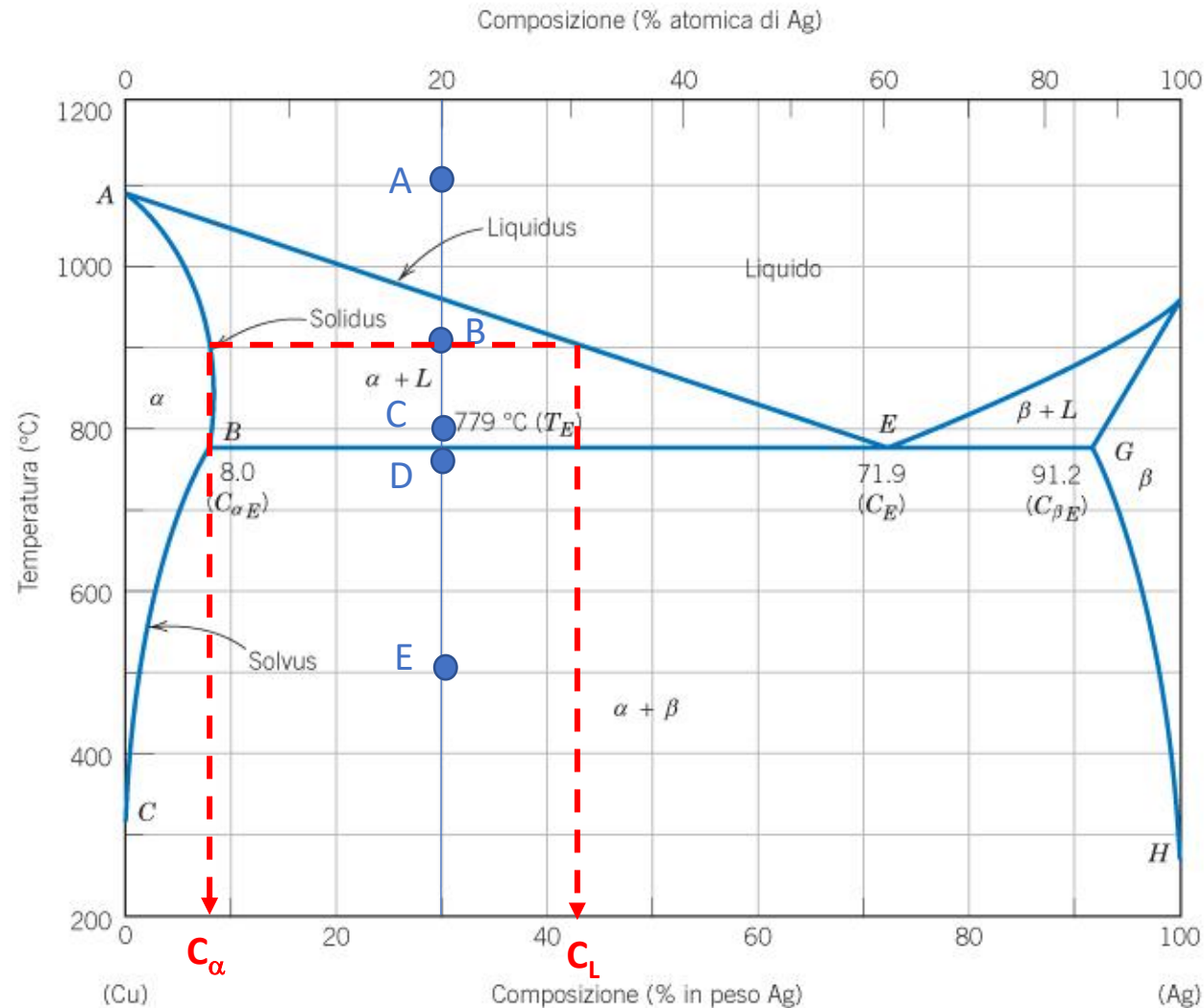
Point A:

Phases: 1 phase liquid

Composition: 30%Ag, 70%Cu

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

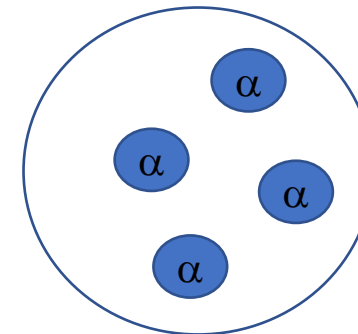
Composition of liquid (L): $C_L = 42\% \text{ Ag}$, $58\% \text{ Cu}$

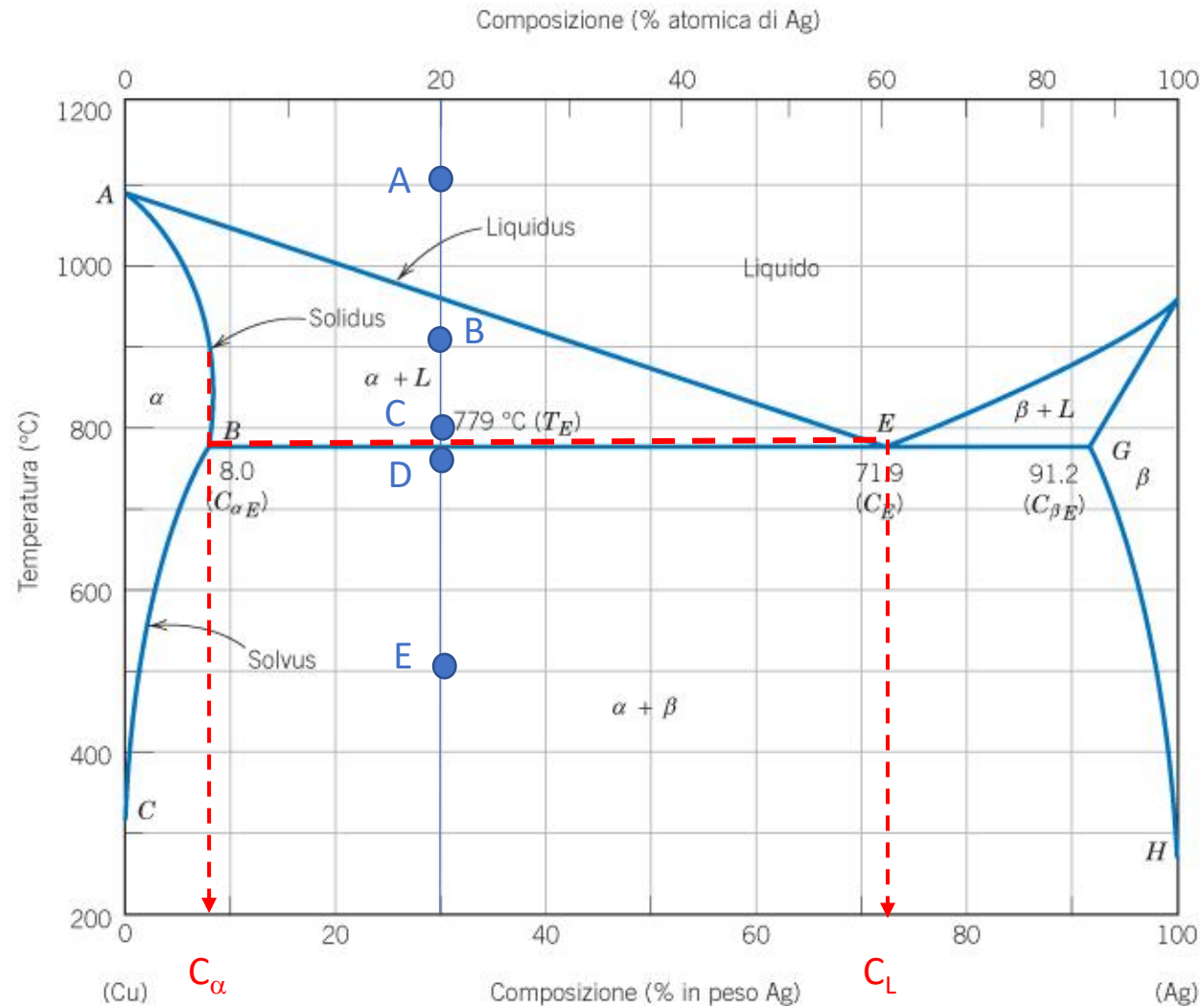
Composition of solid (α): $C_{\alpha} = 8\% \text{ Ag}$, $92\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (42 - 30) / (42 - 8) * 100 = 35\%$$

$$\% L = (30 - 8) / (42 - 8) * 100 = 65\%$$





Point C:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

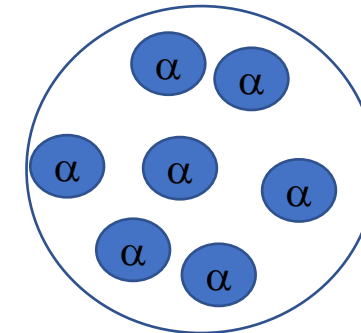
Composition of liquid (L): $C_L = 71.9\% \text{ Ag}$, $28.1\% \text{ Cu}$

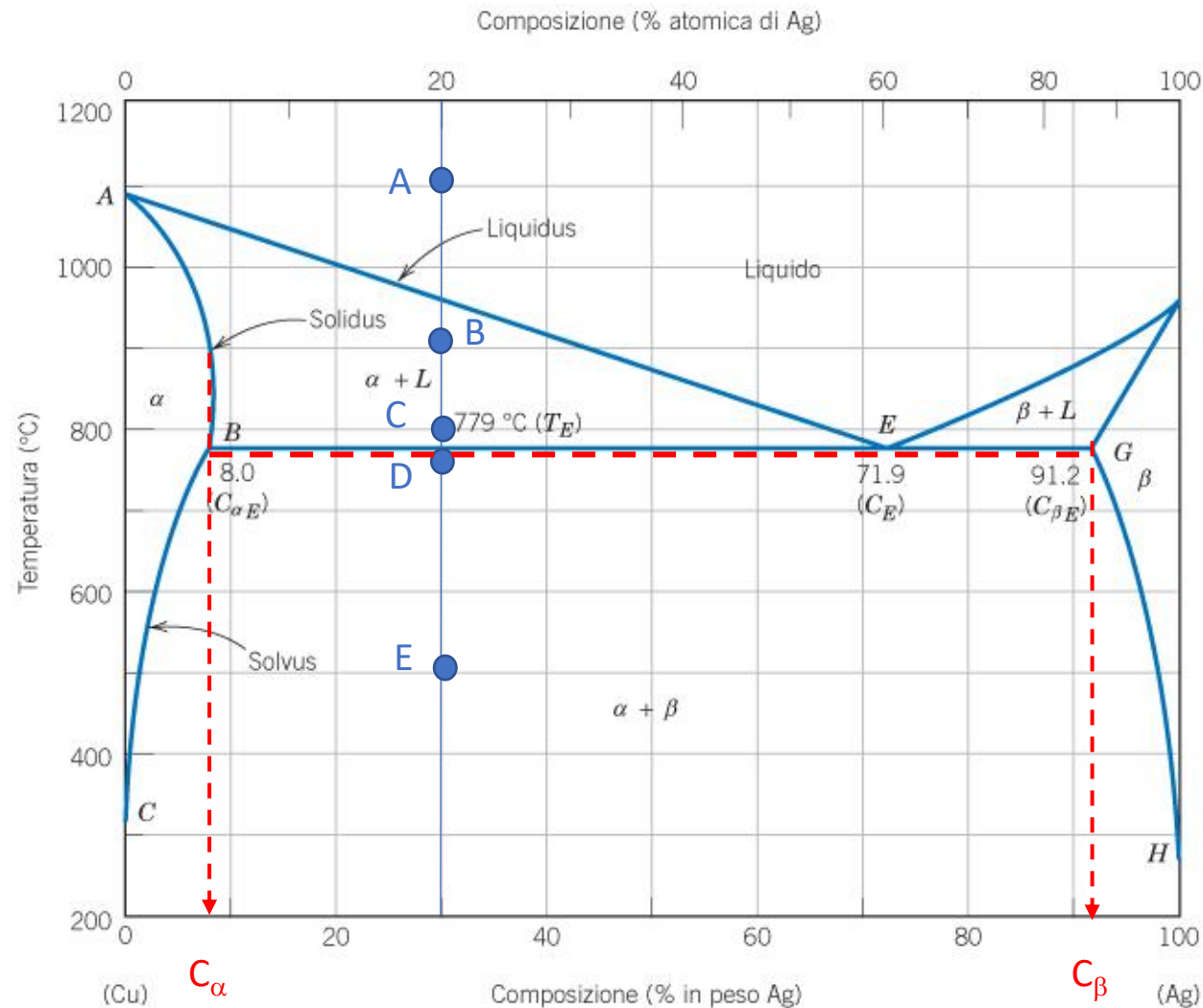
Composition of solid (α): $C_\alpha = 8\% \text{ Ag}$, $92\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (71.9 - 30) / (71.9 - 8) * 100 = 65.6\%$$

$$\% L = (30 - 8) / (71.9 - 8) * 100 = 34.4\%$$





Point D:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

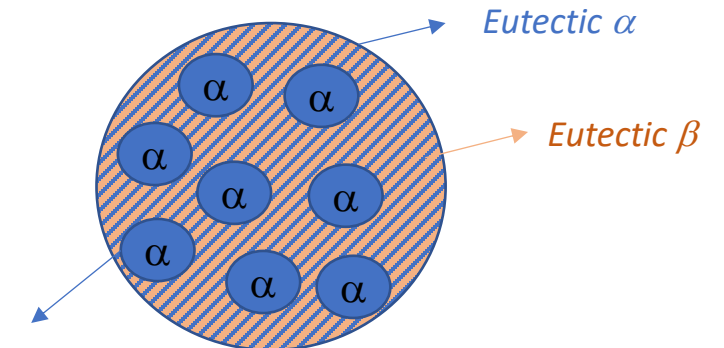
Composition of solid (α): $C_\alpha = 8\% \text{Ag}, 92\% \text{Cu}$

Composition of solid (β): $C_\beta = 91.2\% \text{Ag}, 8.8\% \text{Cu}$

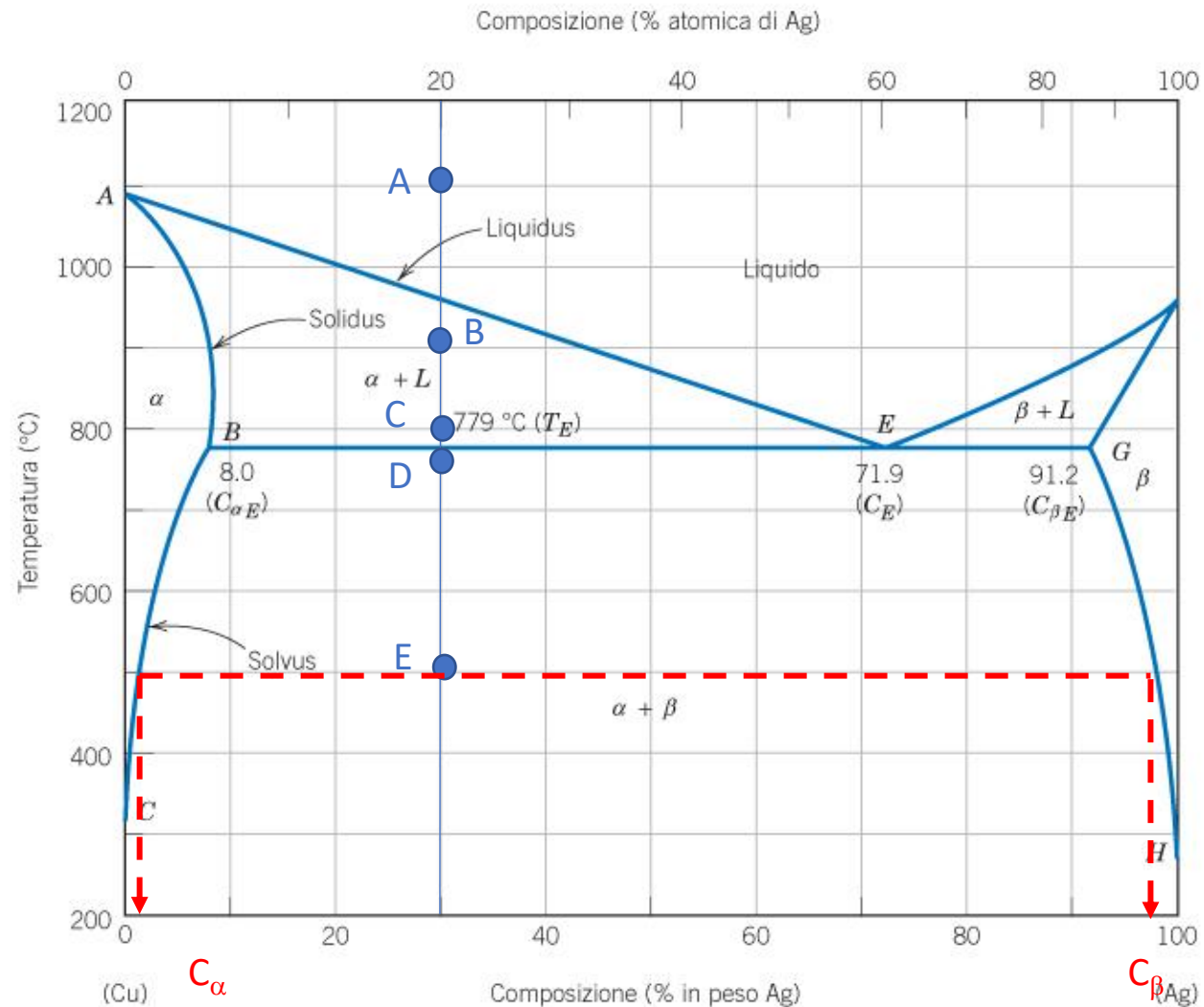
Relative amount: 2 phases region → lever rule

$$\% \alpha = (91.2 - 30) / (91.2 - 8) * 100 = 73.5\%$$

$$\% \beta = (30 - 8) / (91.2 - 8) * 100 = 26.5\%$$



Primary/pro-eutectic α



Point D:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

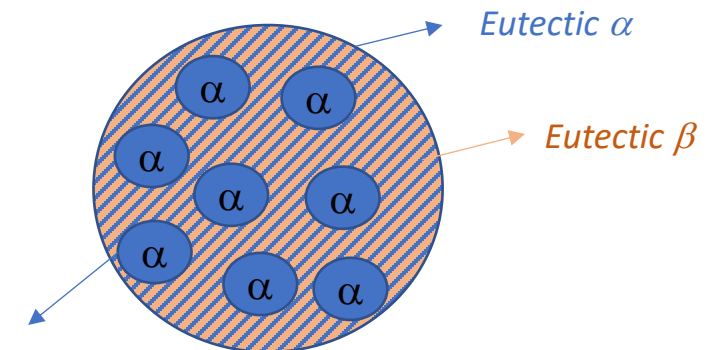
Composition of solid (α): $C_{\alpha} = 2\% \text{Ag}, 98\% \text{Cu}$

Composition of solid (β): $C_{\beta} = 98\% \text{Ag}, 2\% \text{Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (98 - 30) / (98 - 2) * 100 = 71\%$$

$$\% \beta = (30 - 2) / (98 - 2) * 100 = 29\%$$



Primary/pro-eutectic α

Eutectic phase diagram

